9.8 Taylor and Maclaurin series

We know from Section 9.7 that within its interval of convergence I, the sum of a power series is a continuous function with derivatives of all orders. But what about the other way around? If a function $f(x)$ has derivatives of all orders on an interval, can it be expressed as a power series on at least part of that interval? And if it can, what are its coefficients?

If a power series $\sum_{n=0}^{\infty} c_n (x-a)^n$ has a positive radius of convergence R, then the sum of the series defines a function $f(x)$ on the interval $(a - R, a + R)$.

We say that the power series is a ***representation*** of $f(x)$ on that interval.

Question: What is the relationship between the function $f(x)$ and the coefficients $c_0, c_1, c_2, \cdots$ of the power series?

The following theorem answers this question.

Theorem 1: Suppose the power series $\sum_{n=0}^{\infty} c_n (x-a)^n = f(x)$ for $|x - a| < R$, where $R > 0$.

Then $c_k = \dfrac{f^{(k)}(a)}{k!}$ for $k = 0, 1, 2, 3, \cdots$.

$$f(x) = \sum_{n=0}^{\infty} a_n (x - a)^n$$
$$= a_0 + a_1(x - a) + a_2(x - a)^2 + \cdots + a_n(x - a)^n + \cdots$$

with a positive radius of convergence. By repeated term-by-term differentiation within the interval of convergence I, we obtain

$$f'(x) = a_1 + 2a_2(x - a) + 3a_3(x - a)^2 + \cdots + na_n(x - a)^{n-1} + \cdots,$$
$$f''(x) = 1 \cdot 2a_2 + 2 \cdot 3a_3(x - a) + 3 \cdot 4a_4(x - a)^2 + \cdots,$$
$$f'''(x) = 1 \cdot 2 \cdot 3a_3 + 2 \cdot 3 \cdot 4a_4(x - a) + 3 \cdot 4 \cdot 5a_5(x - a)^2 + \cdots,$$

with the nth derivative being

$$f^{(n)}(x) = n!a_n + \text{a sum of terms with } (x - a) \text{ as a factor.}$$

Since these equations all hold at $x = a$, we have

$$f'(a) = a_1, \qquad f''(a) = 1 \cdot 2a_2, \qquad f'''(a) = 1 \cdot 2 \cdot 3a_3,$$

and, in general,

$$f^{(n)}(a) = n!a_n.$$
$$a_n = \frac{f^{(n)}(a)}{n!}.$$

If f has a series representation, then the series must be

$$f(x) = f(a) + f'(a)(x - a) + \frac{f''(a)}{2!}(x - a)^2$$
$$+ \cdots + \frac{f^{(n)}(a)}{n!}(x - a)^n + \cdots. \qquad (1)$$

Definition 1: Let f be a function with derivatives of all orders throughout some interval containing a as an interior point. Then the **Taylor series generated by f at $x = a$** is

$$\sum_{k=0}^{\infty} \frac{f^{(k)}(a)}{k!}(x-a)^k = f(a) + f'(a)(x-a) + \frac{f''(a)}{2!}(x-a)^2 + \cdots + \frac{f^{(n)}(a)}{n!}(x-a)^n + \cdots.$$

The **Maclaurin series of f** is the Taylor series generated by f at $x = 0$, or

$$\sum_{k=0}^{\infty} \frac{f^{(k)}(0)}{k!}x^k = f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 + \cdots + \frac{f^{(n)}(0)}{n!}x^n + \cdots.$$

Example 1: Find the

(a) Taylor series generated by $f(x) = \sin x$ at the point $x = \pi$.

(b) Taylor polynomial of order 5 generated $f(x) = \sin x$ at the point $x = \pi/2$.

(c) Maclaurin series generated by $f(x) = \sin x$.

(d) Maclaurin series generated by $g(x) = \sin 3x^2$.

Solution:

$f^{(k)}(x)$	(a) $\frac{f^{(k)}(\pi)}{k!}$	(b) $\frac{f^{(k)}(\pi/2)}{k!}$	(c) $\frac{f^{(k)}(0)}{k!}$
$f(x) = \sin x$	0	1	0
$f'(x) = \cos x$	-1	0	1
$f''(x) = -\sin x$	0	$-\frac{1}{2!}$	0
$f'''(x) = -\cos x$	$\frac{1}{3!}$	0	$-\frac{1}{3!}$
$f^{(4)}(x) = \sin x$	0	$\frac{1}{4!}$	0
$f^{(5)}(x) = \cos x$	$-\frac{1}{5!}$	0	$\frac{1}{5!}$
$\vdots$	$\vdots$	$\vdots$	$\vdots$
$f^{(2k)}(x) = (-1)^k \sin x$	0	$\frac{(-1)^k}{(2k)!}$	0
$f^{(2k+1)}(x) = (-1)^k \cos x$	$\frac{(-1)^{k+1}}{(2k+1)!}$	0	$\frac{(-1)^k}{(2k+1)!}$
$\vdots$	$\vdots$	$\vdots$	$\vdots$

(a) The Taylor series generated by $f(x) = \sin x$ at the point $x = \pi$ is

$$\sum_{n=0}^{\infty} \frac{f^{(n)}(\pi)}{n!}(x-\pi)^n = -(x-\pi) + \frac{(x-\pi)^3}{3!} - \frac{(x-\pi)^5}{5!} + \cdots + \frac{(-1)^{k+1}}{(2k+1)!}(x-\pi)^{2k+1} + \cdots$$

$$= \sum_{n=0}^{\infty} \frac{(-1)^{n+1}}{(2n+1)!}(x-\pi)^{2n+1}.$$

(b) The Taylor polynomial of order 5 generated by $f(x) = \sin x$ at the point $x = \pi/2$ is $P_5(x) = \sum_{n=0}^{5} \frac{f^{(n)}(\pi/2)}{n!}\left(x - \frac{\pi}{2}\right)^n = 1 - \frac{1}{2}\left(x - \frac{\pi}{2}\right)^2 + \frac{1}{24}\left(x - \frac{\pi}{2}\right)^4$.

(Take note that this 5-th order Taylor polynomial has degree 4.)

(c) The Maclaurin series generated by $f(x) = \sin x$ is

$$\sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!}x^n = x - \frac{x^3}{3!} + \frac{x^5}{5!} + \cdots + \frac{(-1)^k x^{2k+1}}{(2k+1)!} + \cdots = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)!}x^{2n+1}.$$

(d) The Maclaurin series generated by $f(x) = \sin 3x^2$ is

$$\sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)!}(3x^2)^{2n+1} = 3x^2 - \frac{(3x^2)^3}{3!} + \frac{(3x^2)^5}{5!} + \cdots + \frac{(-1)^k (3x^2)^{2k+1}}{(2k+1)!} + \cdots$$

$$= 3x^2 - \frac{9x^6}{2} + \frac{81x^{10}}{40} + \cdots + \frac{(-1)^k 3^{2k+1} x^{4k+2}}{(2k+1)!} + \cdots.$$

Example 2:

Find the Maclaurin series for the function $f(x) = \ln(1-x)$ for $-1 \le x < 1$.

Solution:

$f^{(k)}(x)$	$\frac{f^{(k)}(0)}{k!}$
$f(x) = \ln(1-x)$	0
$f'(x) = \frac{-1}{1-x}$	-1
$f''(x) = \frac{1}{(1-x)^2}$	$\frac{1}{2}$
$f'''(x) = \frac{-2}{(1-x)^3}$	$-\frac{2}{3!} = -\frac{1}{3}$
$f^{(4)}(x) = \frac{2(3)}{(1-x)^4}$	$\frac{6}{4!} = \frac{1}{4}$
$f^{(5)}(x) = \frac{-2(3)(4)}{(1-x)^5}$	$-\frac{24}{5!} = -\frac{1}{5}$
$\vdots$	$\vdots$
$f^{(k)}(x) = (-1)^k \frac{(k-1)!}{(1-x)^k}$	$(-1)^k \frac{(k-1)!}{k!} = \frac{(-1)^k}{k}$
$\vdots$	$\vdots$

So, the Maclaurin series for $f(x) = \ln(1+x)$ is

$$\ln(1+x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!} x^n = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \cdots + \frac{(-1)^{k-1}}{k} x^k + \cdots = \sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n} x^n .$$

Definition 2: Let f be a function with derivatives of order k for $k = 1, 2, \ldots, N$ in some interval containing a as an interior point. Then for any integer n from 0 through N, the **Taylor polynomial of order n** generated by f at $x = a$ is the polynomial

$$P_n(x) = f(a) + f'(a)(x-a) + \frac{f''(a)}{2!}(x-a)^2 + \cdots + \frac{f^{(k)}(a)}{k!}(x-a)^k + \cdots + \frac{f^{(n)}(a)}{n!}(x-a)^n.$$

Example 3: Find the Taylor series and the Taylor polynomials generated by $f(x) = e^x$ at $x = 0$.

Maclaurin series

Solution Since $f^{(n)}(x) = e^x$ and $f^{(n)}(0) = 1$ for every $n = 0, 1, 2, \ldots$, the Taylor series generated by f at $x = 0$ (see Figure 9.22) is

$$f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 + \cdots + \frac{f^{(n)}(0)}{n!}x^n + \cdots$$

$$= 1 + x + \frac{x^2}{2} + \cdots + \frac{x^n}{n!} + \cdots$$

$$= \sum_{k=0}^{\infty} \frac{x^k}{k!}.$$

This is also the Maclaurin series for e^x. In the next section we will see that the series converges to e^x at every x.

The Taylor polynomial of order n at $x = 0$ is

$$P_n(x) = 1 + x + \frac{x^2}{2} + \cdots + \frac{x^n}{n!}.$$

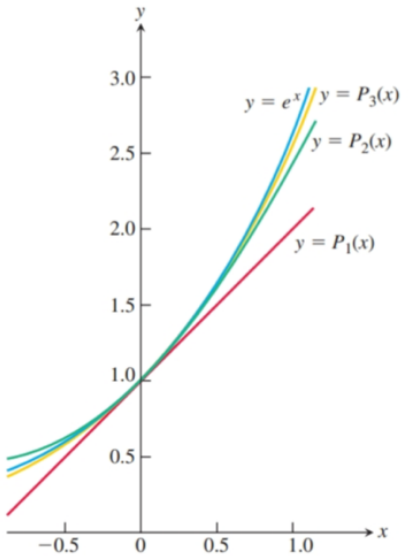


Figure 1 The graph of $f(x) = e^x$ and its Taylor polynomials

$P_1(x) = 1 + x$

$P_2(x) = 1 + x + (x^2/2!)$

$P_3(x) = 1 + x + (x^2/2!) + (x^3/3!)$.

Notice the very close agreement near the center $x = 0$ (Example 3).

Example 4: Find the Taylor series and Taylor polynomials generated by $f(x) = \cos x$ at $x = 0$.

Solution The cosine and its derivatives are

$$\begin{aligned} f(x) &= \cos x, & f'(x) &= -\sin x, \\ f''(x) &= -\cos x, & f^{(3)}(x) &= \sin x, \\ &\vdots & &\vdots \\ f^{(2n)}(x) &= (-1)^n \cos x, & f^{(2n+1)}(x) &= (-1)^{n+1} \sin x. \end{aligned}$$

At $x = 0$, the cosines are 1 and the sines are 0, so

$$f^{(2n)}(0) = (-1)^n, \qquad f^{(2n+1)}(0) = 0.$$

The Taylor series generated by f at 0 is

$$\begin{aligned} &f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 + \frac{f'''(0)}{3!}x^3 + \cdots + \frac{f^{(n)}(0)}{n!}x^n + \cdots \\ &\quad = 1 + 0 \cdot x - \frac{x^2}{2!} + 0 \cdot x^3 + \frac{x^4}{4!} + \cdots + (-1)^n \frac{x^{2n}}{(2n)!} + \cdots \\ &\quad = \sum_{k=0}^{\infty} \frac{(-1)^k x^{2k}}{(2k)!}. \end{aligned}$$

This is also the Maclaurin series for $\cos x$. Notice that only even powers of x occur in the Taylor series generated by the cosine function, which is consistent with the fact that it is an even function. In Section 9.9, we will see that the series converges to $\cos x$ at every x.

Because $f^{(2n+1)}(0) = 0$, the Taylor polynomials of orders $2n$ and $2n + 1$ are identical:

$$P_{2n}(x) = P_{2n+1}(x) = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \cdots + (-1)^n \frac{x^{2n}}{(2n)!}.$$

Figure 2 shows how well these polynomials approximate $f(x) = \cos x$ near $x = 0$. Only the right-hand portions of the graphs are given because the graphs are symmetric about the y-axis. ■

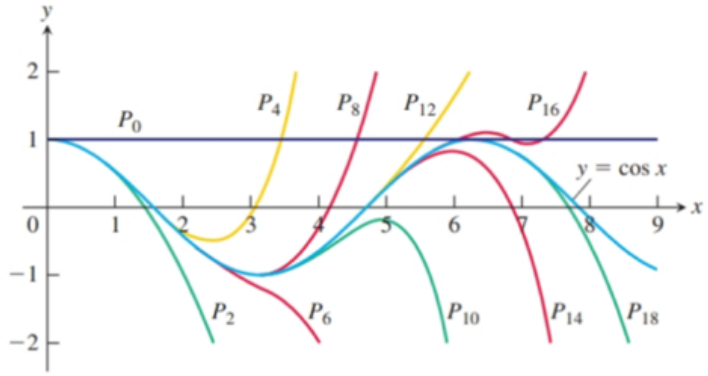


Figure 2 The polynomials

$$P_{2n}(x) = \sum_{k=0}^{n} \frac{(-1)^k x^{2k}}{(2k)!}$$

converge to $\cos x$ as $n \to \infty$. We can deduce the behavior of $\cos x$ arbitrarily far away solely from knowing the values of the cosine and its derivatives at $x = 0$ (Example 4).